

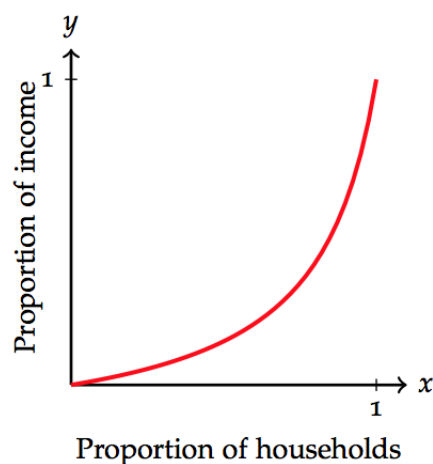
6.1 Income Inequality and the Gini Index

Inequality among a group of people can affect opportunity and access to housing, food, medicine, education, jobs, voting, and other essentials. Income inequality is one of the most easily quantified measures of inequality. In this section we will learn ideas related to income inequality within a country's borders.

Lorenz Curve

Let P designate the set of _____ in a country during a given period of time.

The Lorenz curve, $L_P(x)$, for that country and year contains the points (x, y) , such that the _____ $(100x)\%$ of households received $(100y)\%$ of the country's total income.



Properties of the Lorenz Curve

For any population set P :

- The **domain** of the Lorenz curve is the closed interval _____, representing 0% - 100% of the households under consideration.
- The **range** of the Lorenz curve is also the closed interval _____, representing 0% - 100% of the income under consideration.
- $L_P(0) = 0$.
- $L_P(1) = 1$.
- L_P is an _____ function.
This is because x represents the poorest $(100x)\%$ of households, so households to the right of x receive more income.
- $L_P(x) \leq$ _____.
This means that the poorest $(100x)\%$ of households cannot receive more than $(100x)\%$ of the income (otherwise, they wouldn't be the poorest households).

Example 1

Suppose $L_P(x) = 0.3x + 0.7x^5$ defines a Lorenz curve for a population set P .

(a) Interpret in words the statement $L_P(0.45) = 0.148$.

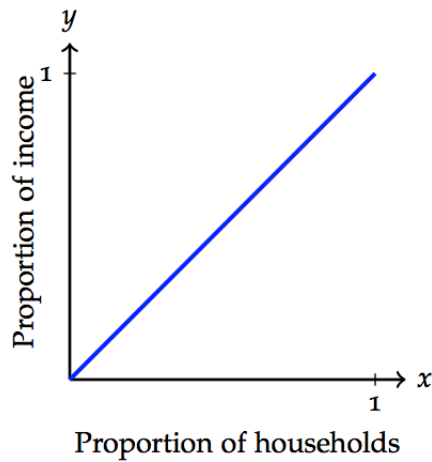
(b) How much income was earned by the poorest 20% of households in P ?

(c) How much income was earned by the wealthiest 10% of households in P ?

Perfectly Equal Income Distribution

Let Id represent a hypothetical egalitarian country in which every household earns an equal share of income.

So $L_{Id}(x) =$ is the Lorenz curve for this population.

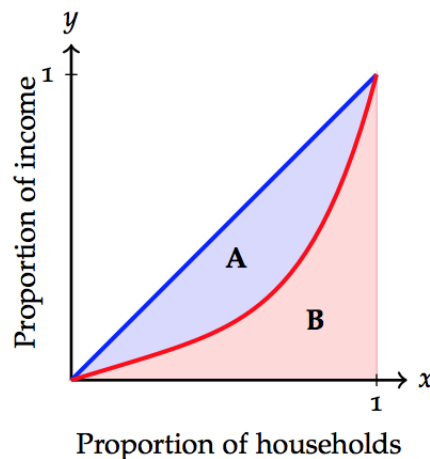


Based on this Lorenz curve, we see for example that the poorest 20% of the population received 20% of the income, while the wealthiest 20% of the population

The Gini Index

The Gini index represents the size of the gap between a country's Lorenz curve and the egalitarian Lorenz curve as a proportion of the total area under the egalitarian Lorenz curve, and can be computed using the formula:

$$G = \frac{\int_0^1 (L_{Id}(x) - L_P(x)) dx}{\int_0^1 L_{Id}(x) dx} = \frac{\int_0^1 (x - L_P(x)) dx}{\int_0^1 x dx} = 1 - 2 \int_0^1 L_P(x) dx$$



The Gini index corresponding to a Lorenz curve is the ratio of the area of the gap between the Lorenz curve and the egalitarian curve

(_____) to the whole triangle (_____).

This index can range from _____ (every household receives the same income) to _____ (a single totalitarian household receives all the income in

the entire country).

Properties of the Gini Index

- The Gini index can range from a low of 0 (every household receives equal income) to a high of 1 (a single household receives all of the income and every other household receives nothing).
- A higher index indicates that the country's income distribution is more _____.
- The Gini index is often stated as a percentage, e.g. $G = 0.3$ would be reported as _____.
- World Bank estimates for the Gini indices of actual countries range from 0.255 to 0.630. CIA estimates, which use a different methodology, range from 0.232 to 0.632.
- Taxation, which funds programs for lower-income people, can change a country's Gini Index. OECD (The Organisation for Economic Co-operation and Development) estimates the United States' pre-tax Gini index at 0.486, and its post-tax Gini index at 0.378.

Example 2

(a) Compute the Gini index for the Lorenz curve $L_P(x) = 0.5x^2 + 0.5x^4$.

(b) The Gini index for the Lorenz curve $L_P(x) = x^a$ is $G = 0.8$. Find the value of a .

Approximating the Gini Index

In practice, it is not possible to obtain a precise equation for a country's Lorenz curve. In almost all cases, we get a table reflecting careful estimation of the Lorenz curve value at selected percentages.

The table below shows Lorenz curve data for several countries, as estimated by the World Bank. Each estimate was based on that country's most recent census, and all are within the past ten years.

	Proportion of households					
	0	0.2	0.4	0.6	0.8	1
Brazil	0	0.029	0.100	0.224	0.414	1
China	0	0.047	0.144	0.297	0.529	1
India	0	0.085	0.206	0.363	0.571	1
Mexico	0	0.049	0.137	0.270	0.472	1
Nigeria	0	0.044	0.127	0.257	0.460	1
Russia	0	0.061	0.165	0.313	0.526	1

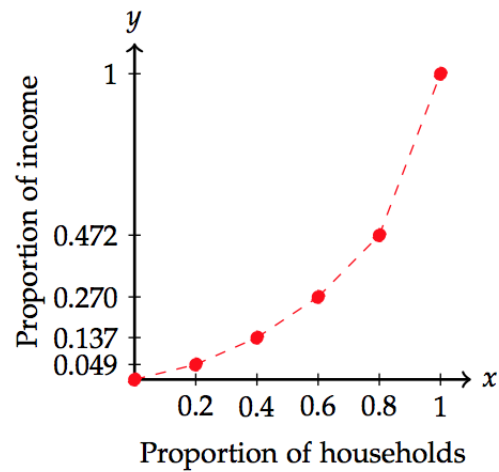
Some reasons why a precise Lorenz curve cannot be obtained:

- The income of agricultural workers depends on unpredictable factors like the weather, and varies from season to season.
- Workers who earn tips or commissions do not earn the same amounts consistently.
- Workers in illicit trades, like drugs or sex work, do not report their true income.

Example 3

For this example we will use the World Bank tabular data from the previous page.

- (a) Use the Trapezoidal rule to approximate the Gini index for Mexico.



- (b) Why can't we use Simpson's rule to approximate the Gini index for any country in the table given?

(c) Use the Trapezoidal rule to approximate the Gini index for Russia.

(d) Using the Gini indices calculated above for Mexico and Russia, deduce in which country the income is more equally distributed.